The empirical recurrence for OEIS A224562, proved by transfer matrix

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Abstract

OEIS A224562 counts the $(n+3)\times 5$ matrices over $\{0,\dots,1\}$ all of whose 4×4 contiguous subblocks are idempotent, and carries an empirical recurrence of order 12 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition ties together 4 consecutive rows and 4 consecutive columns at once, so a single row is not a state; the strip of the last 3 rows is, and the admissible strips are read off the overlap graph of the admissible windows instead of being enumerated. That makes $a(n)$ a walk count on $S = 366$ vertices, hence C-finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation.

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1 The conjecture

OEIS A224562 is “Number of (n+3) X 5 0..1 matrices with each 4 X 4 subblock idempotent.”. It has offset 1 and begins

$$370, 340, 428, 517, 611, 775, 1044, 1383, \ \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 2*a(n-1) - a(n-2) + a(n-3) - 2*a(n-5) + a(n-6) - 2*a(n-7) + 2*a(n-8) + a(n-11) - a(n-12) for n>15.

As of the “Last modified” line on the live entry (Sep 01 2018 09:17:36, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

Call a 4×4 matrix over $\{0,\dots,1\}$ a *window* when it is idempotent, and call a 4×5 matrix a *band* when every one of its 2 contiguous 4×4 subblocks is a window. What the entry counts is the matrices of 5 columns whose every 4 consecutive rows form a band.

Take as states the 3×5 strips that occur as the top 3 rows of a band, together with those that occur as its bottom 3 rows, and put an edge from u to v when the band whose first 3 rows are u and whose last 3 rows are v exists. A band is determined by that pair, so the edge is unique when it exists. Reading a counted matrix from the top, its rows 1..3, then 2..4, and so on, form a walk, and every walk arises from exactly one matrix; a matrix with $n+3$ rows gives a walk of n edges. Since the first 3 rows of a counted matrix with at least 4 rows are the top of a band, and since no further condition falls on where a walk starts or ends,

$$a(n) \;=\; \iota^{\top} M^{n} \tau, \qquad \iota = \tau = (1, 1, \dots, 1)^{\top},$$

with M the adjacency matrix on $S = 366$ states. In particular a is C-finite. The alignment was checked against the entry's published terms rather than assumed.

There are $(1+1)^{15}$ strips of 3 rows and 5 columns, far too many to test one by one, and they are not tested one by one. Two consecutive windows of a band overlap in 4×3 entries, so a band is exactly a walk of 2 windows in the graph that joins a window to those whose first 3 columns are its last 3 columns. Enumerating those walks produces the bands, and with them the states and the edges, directly.

3 The admissible windows

The vertices of the overlap graph are the 0/1 matrices B of size 4×4 with $B^2 = B$. There are 2^{16} matrices to sift, which is already awkward at $4 = 4$ and hopeless a little beyond it, so they are not sifted; they are generated from the following description, which also makes the count itself transparent.

Lemma 1. *Let B be a $k \times k$ matrix with entries in $\{0,1\}$, let r_i denote its ith row and $S_i = \{j : B_{ij} = 1\}$ the support of that row. Then $B^2 = B$ if and only if*

$$r_i = \sum_{t \in S_i} r_t \qquad (1 \le i \le k),$$

the rows on the right necessarily having pairwise disjoint supports. Writing $Z = \{i : r_i = 0\}$ and $U = \{1, \dots, k\} \setminus Z$, and calling $i \in U$ primitive *when $S_i \cap U = \{i\}$, the idempotent 0/1 matrices are exactly those obtained as follows: choose Z; choose the set $B^* \subseteq U$ of primitive indices and for each $b \in B^*$ a set $Z_b \subseteq Z$, putting $r_b = e_b + \sum_{j \in Z_b} e_j$; and for each $i \in U \setminus B^*$ choose a nonempty $B_i \subseteq B^*$ whose rows have pairwise disjoint supports, putting $r_i = \sum_{b \in B_i} r_b$. Distinct data give distinct matrices.*

Proof. The ith row of B^2 is $\sum_t B_{it} r_t = \sum_{t \in S_i} r_t$, so $B^2 = B$ is precisely the displayed system. Its entries lie in $\{0,1\}$, so for each i the supports S_t, $t \in S_i$, are pairwise disjoint with union S_i.

Assume $B^2 = B$. Rows indexed by Z vanish, so for $i \in U$ we may write $r_i = \sum_{t \in S_i \cap U} r_t$ with $S_i \cap U \neq \emptyset$. If i is primitive this says $S_i \subseteq \{i\} \cup Z$, which is the stated shape with $Z_i = S_i \cap Z$. For the rest argue by induction on $|S_i|$. The supports S_t, $t \in S_i \cap U$, are disjoint, nonempty and cover S_i; if one of them has $|S_t| = |S_i|$ it is the only one, whence $r_t = r_i$, $S_t \cap U = S_i \cap U = \{t\}$ and t is primitive. Otherwise every such t has $|S_t| < |S_i|$, and each r_t is by induction a sum of primitive rows with disjoint supports. Either way $r_i = \sum_{b \in B_i} r_b$ with $B_i \subseteq B^*$, and intersecting $S_i = \bigsqcup_{b \in B_i} (\{b\} \cup Z_b)$ with U returns B_i, so the data is recovered from B and distinct data give distinct matrices.

Conversely, given data of the stated shape, the disjointness makes every entry 0 or 1, and $\sum_{t \in S_i} r_t$ equals 0, r_b, $\sum_{b \in B_i} r_b$ in the three cases, which is r_i each time. Hence $B^2 = B$. □

Generating from Lemma 1 gives 452 admissible 4×4 windows. For $k = 2, 3, 4$ the same generation gives 8, 50 and 452, which agree with exhaustion over all 2^{k^2} matrices; that is the check that the description above is neither too wide nor too narrow.

4 The proof

Lemma 2. *Let $q(t) = t^{12} - \sum_i c_i t^{12-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+12} - \sum_i c_i M^{j+12-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) = 2\,a(n-1) - a(n-2) + a(n-3) - 2\,a(n-5) + a(n-6) - 2\,a(n-7) + 2\,a(n-8) + a(n-11) - a(n-12)$$

for every $n > 15$.

Proof. The vector $w = q(M)\tau$ was formed by 12 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 15 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The bound is exact: at $n = 15$ the recurrence fails on the entry's own published terms, so no larger range holds. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 35 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A224562.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] K. H. Kim, *Boolean Matrix Theory and Applications*, Marcel Dekker, 1982.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.